

Licensed Software Installation Request Management in ServiceNow

ServiceNow Custom Application

Project	Details
Project	Licensed Software Installation Request
Platform	ServiceNow
Interface	Service Portal / Service Catalog
Workflow	Catalog → REQ/RITM → License Validation → Approval → Task → Fulfillment → Notification

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1. Project Overview

The Licensed Software Installation Request project provides a structured and automated way for employees to request installation of licensed software through the ServiceNow Service Catalog. The solution manages the request from submission through license validation, approval, fulfillment, notification, and closure.

Employees can submit licensed software requests through the Service Catalog.

Required software, employee, device, business, urgency, and license information is captured.

ServiceNow creates Request and Requested Item records.

The workflow retrieves catalog variables and checks license availability.

Approved requests generate a Catalog Task for the Software Support Team.

Notifications and SLAs provide process visibility and time-based monitoring.

Dashboard and Update Set configuration support reporting and deployment.

2. Problem Statement and Objectives

Software installation requests can become difficult to control when they are submitted through unstructured channels. This project provides a centralized ServiceNow process for collecting complete request information, validating licensing, obtaining approval, creating fulfillment tasks, and communicating progress.

Objectives

Create a centralized Licensed Software Installation Request Catalog Item.

Capture complete request information using structured catalog variables.

Automate license validation and approval.

Create software installation tasks automatically after approval.

Route fulfillment work to the Software Support Team.

- Monitor processing time using SLAs.
- Send notifications at important stages.
- Provide dashboard visibility and controlled deployment through Update Sets.

3. Application and Data Architecture

The solution is developed on ServiceNow using Service Catalog, Requested Item records, workflow automation, approvals, Catalog Tasks, notifications, SLAs, dashboard reporting, and security controls.

Application flow:

Catalog Item → REQ/RITM → Catalog Variables → License Validation → Approval → Software Asset Approval → Catalog Task → Fulfillment → Notification → Closure

Component	Purpose
Service Catalog	Employee-facing request entry point.
Catalog Variables	Capture software and installation requirements.
UI Policies / Client Scripts	Control form behavior and validation.
Workflow / Flow Designer	Automate processing and decisions.
License Validation	Check requested software licensing.
Approval	Control whether request may proceed.
Catalog Task	Provide fulfillment work item.
Assignment Group	Route task to Software Support Team.
Notifications	Communicate key events.
Dashboard	Provide operational visibility.
Update Set	Package configuration for deployment.

4. Requirement Analysis and Planning

The project requirements were identified to automate the complete licensed software installation process.

Functional Requirements

- Users submit requests through the Service Catalog.
- Required details are captured using catalog variables.
- REQ and RITM records are created.
- License availability is validated.
- Approval is obtained before fulfillment.
- Approved requests create a Catalog Task.
- Tasks are assigned to Software Support Team.
- Users receive status and completion notifications.
- SLAs support priority-based processing.
- Access is controlled using roles and ACLs.

Planning covered catalog configuration, form customization, UI logic, workflow automation, approval and license validation, task and notification configuration, testing, security, deployment, and documentation.

5. Catalog Item and Variable Configuration

A dedicated Licensed Software Installation Request Catalog Item was configured to allow employees to submit software installation requests.

Variable	Purpose
Which software do you want?	Requested software.
Which software version you want?	Required version.
Business Justification	Business need.
By which date you need it?	Required installation date.
Enter employee name	Employee identification.
Where do you want to install your software/device?	Target device.
Enter your manager name	Approval routing.
Enter your department name	Department.
Urgency	Priority and SLA path.
Select License Type	Licensing category.

These variables ensure complete and accurate information is available for downstream validation, approval, and fulfillment.

The screenshot displays the ServiceNow interface for a 'Software Installation Request'. The breadcrumb trail is 'Home > Service Catalog > Software > Software Installation Request'. The form title is 'Software Installation Request' with the subtitle 'Request to install company-approved software'. A description states: 'Allows employees to request software installations. Fulfills via Software Support Team.' The form contains several input fields: 'What software do you need?' with the value 'Microsoft Office 365', 'Specify version (if required)' with the value '2021', 'Why do you need this software?' with the value 'Required for daily work', and 'Select urgency level.' with a dropdown menu set to 'High'. On the right side, there is a 'Quantity' dropdown set to '1', a 'Delivery Time' of '2 Days', and buttons for 'Add to Cart', 'Save as Draft', and 'Order Now'. The bottom of the screen shows a Windows taskbar with the date '21-20 09-09-2026'.

Fig 5.1 – Licensed Software Installation Request – Configured Catalog Variables / Request Details

6. Service Portal Interface and Usability

The Service Portal provides a simple interface for employees to submit licensed software requests. The form groups software, requester, installation, business, urgency, and licensing information so users can provide complete details before submission.

This screenshot shows the same 'Software Installation Request' form from a different perspective, highlighting the user-friendly interface. The breadcrumb trail is 'Home > Service Catalog > Software > Software Installation Request'. The form title is 'Software Installation Request' with the subtitle 'Request to install company-approved software'. The description is 'Allows employees to request software installations. Fulfills via Software Support Team.' The input fields are: 'What software do you need?' (empty), 'Specify version (if required)' (empty), 'Why do you need this software?' (empty), and 'Select urgency level.' with a dropdown menu set to 'Critical'. On the right side, there is a 'Quantity' dropdown set to '1', a 'Delivery Time' of '0 Days', and buttons for 'Add to Cart', 'Save as Draft', and 'Order Now'. The bottom of the screen shows a Windows taskbar with the date '21-20 09-09-2026'.

Fig 6.1 – Licensed Software Installation Request – User-Friendly Catalog Form

7. Workflow Automation and Approval

The SIRA_Software_Installation_Workflow automates catalog-variable retrieval, urgency processing, license lookup, approval, software asset approval, Catalog Task creation, record updates, and notifications.

Workflow:

Request Submission → Get Catalog Variables → Priority Processing → License Lookup → Approval → Asset Approval → Catalog Task → Update Record → Notification

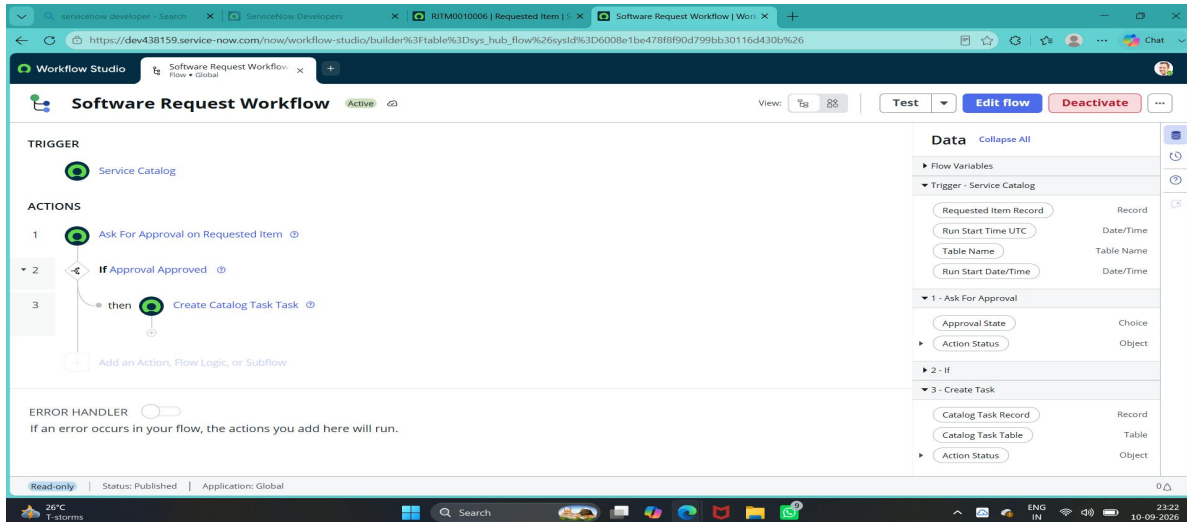


Fig 7.1 – Licensed Software Installation Request – Software Request Workflow

Approval

The request is routed for approval before fulfillment. Approved requests proceed to asset validation and task creation; rejected requests follow the configured rejection and notification path.

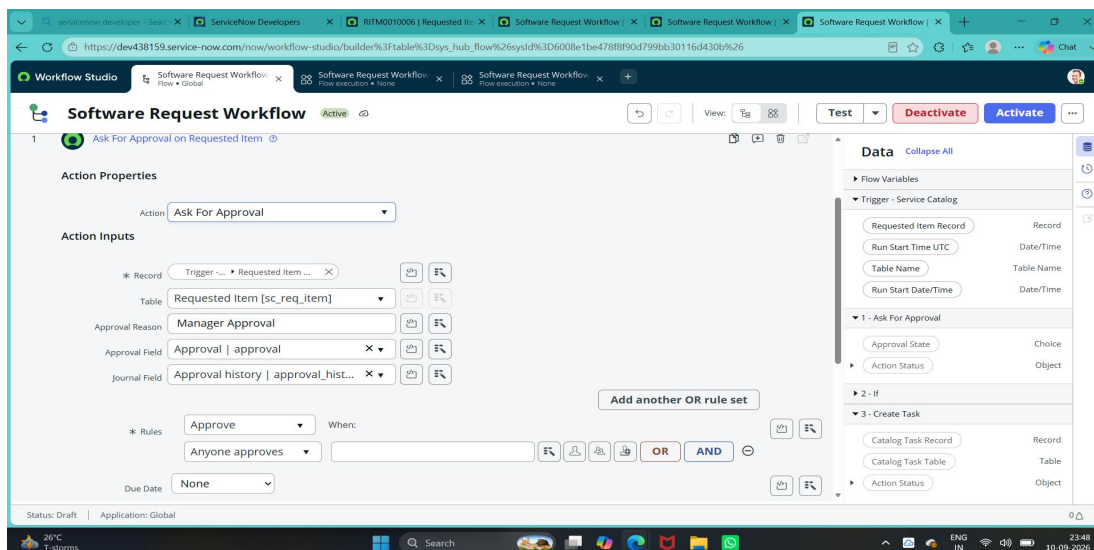


Fig 7.2 – Licensed Software Installation Request – Workflow(Approvals)

8. End-to-End Functional Flow

1. Employee opens the Licensed Software Installation Request.
2. Employee enters catalog variables and submits the request.
3. ServiceNow creates REQ and RITM records.
4. Workflow retrieves catalog variables.
5. Urgency and SLA path are evaluated.
6. License information is checked.
7. Request is routed for approval.
8. Approved request proceeds through software asset approval.
9. Catalog Task is created and assigned to Software Support Team.
10. Support team installs the software.
11. Task and request are updated to completion.
12. Requester receives notifications.

9. Testing and Security Validation

Testing verifies the complete request process from catalog submission through fulfillment and closure.

Test Area	Expected Result
Catalog submission	Complete request submits successfully.
Mandatory fields	Incomplete requests are prevented.
REQ/RITM creation	Records are created correctly.
License validation	License information is checked.
Approval	Correct approval stage is reached.
Task creation	Approved request creates Catalog Task.
Assignment	Task reaches Software Support Team.
Notifications	Configured notifications are generated.
Completion	Fulfillment is recorded.

10. Deployment, Documentation and Final Presentation

The completed configuration is captured in the SIRA_Workflow_Automation Update Set for controlled tracking and movement between ServiceNow instances.

Name	Application	State	Installed from	Created	Created by	Parent	Batch Base
Default	Global	In progress		2026-04-30 00:40:06	system	(empty)	(empty)
Licensed Software Installation - Capstone	Global	Complete		2026-09-02 01:54:28	admin	(empty)	(empty)

Related Links

[Merge Update Sets](#)

Fig 10.1 – Licensed Software Installation Request – Update Set completed

The screenshot shows the ServiceNow interface for Retrieved Update Sets. The browser address bar displays the URL: `dev270509.service-now.com/now/nav/ui/classic/params/target/sys_remote_update_set_list.do%3Fsysparm_fixed_query%3Dsys_class_name%253Dsys_remote_update_set`. The page title is "Retrieved Update Sets". Below the title bar, there is a search bar and a dropdown menu for "Actions on selected rows...". The main table lists the update sets with columns: Name, Application, State, Update source, Description, Loaded, Committed, Parent, and Remote Batch Base. The table contains one row: "Licensed Software Installation - Capstone" with Application "Global", State "Committed", Update source "(empty)", Description "Update Set for Employee Request Installa...", Loaded "2026-09-17 11:24:06", Committed "2026-09-17 23:56:51", Parent "(empty)", and Remote Batch Base "(empty)". Below the table, there is a "Related Links" section with a link "Import Update Set from XML". The bottom of the screenshot shows a Windows taskbar with the date and time "12:28 18-09-2026".

Name	Application	State	Update source	Description	Loaded	Committed	Parent	Remote Batch Base
Licensed Software Installation - Capstone	Global	Committed	(empty)	Update Set for Employee Request Installa...	2026-09-17 11:24:06	2026-09-17 23:56:51	(empty)	(empty)

Fig 10.2 – Retrieved / Committed Update Set Evidence

The screenshot shows the ServiceNow interface for Retrieved Update Set - Licensed Software Installation - Capstone. The browser address bar displays the URL: `dev270509.service-now.com/now/nav/ui/classic/params/target/sys_remote_update_set.do%3Fsys_id%3D6a3f294f47dfc310d799bb30116d43a9%26sysparm_record_target%3Dsys_remote_update_set%2...`. The page title is "Retrieved Update Set - Licensed Software Installation...". Below the title bar, there is a search bar and a dropdown menu for "Actions on selected rows...". The main table lists the update sets with columns: Name, Type, Target name, Table, View, and Action. The table contains 15 rows of data. The bottom of the screenshot shows a Windows taskbar with the date and time "12:33 18-09-2026".

Name	Type	Target name	Table	View	Action
item_option_new_2cf6aa4447930310d799bb30116d4373	Variable	What software do you need?			INSERT_OR_UPDATE
item_option_new_2fb8624847930310d799bb30116d43d1	Variable	Specify version (if required)			INSERT_OR_UPDATE
item_option_new_3529ee4847930310d799bb30116d432d	Variable	Why do you need this software?			INSERT_OR_UPDATE
item_option_new_9eb92e8847930310d799bb30116d43e7	Variable	Select urgency level.			INSERT_OR_UPDATE
open_nlu_driver_language_29267b7d47d78f10d799bb30116d436b	Open NLU Driver Language	Danish			INSERT_OR_UPDATE
open_nlu_driver_language_3c263b7d47d78f10d799bb30116d43f9	Open NLU Driver Language	Canadian French			INSERT_OR_UPDATE
open_nlu_driver_language_3e26bb7d47d78f10d799bb30116d43a0	Open NLU Driver Language	German			INSERT_OR_UPDATE
open_nlu_driver_language_4a26bb7d47d78f10d799bb30116d4302	Open NLU Driver Language	English			INSERT_OR_UPDATE
open_nlu_driver_language_5626bb7d47d78f10d799bb30116d4354	Open NLU Driver Language	Finnish			INSERT_OR_UPDATE
open_nlu_driver_language_5f26bb7d47d78f10d799bb30116d4337	Open NLU Driver Language	Japanese			INSERT_OR_UPDATE
open_nlu_driver_language_6836bb7d47d78f10d799bb30116d43e9	Open NLU Driver Language	Portuguese			INSERT_OR_UPDATE
open_nlu_driver_language_7436bb7d47d78f10d799bb30116d433b	Open NLU Driver Language	Spanish			INSERT_OR_UPDATE
open_nlu_driver_language_8036bb7d47d78f10d799bb30116d439d	Open NLU Driver Language	Norwegian			INSERT_OR_UPDATE
open_nlu_driver_language_8d26bb7d47d78f10d799bb30116d4300	Open NLU Driver Language	Chinese (Simplified)			INSERT_OR_UPDATE
open_nlu_driver_language_8f26bb7d47d78f10d799bb30116d43a7	Open NLU Driver Language	Italian			INSERT_OR_UPDATE

Fig 10.3 – Licensed Software Installation Request-Customer Updates

Documentation

The documentation records the project requirement, architecture, catalog configuration, UI logic, workflow, approval, license validation, task fulfillment, notifications, SLA, testing, security, deployment, and final outcome.

Final Presentation

The final presentation demonstrates the complete lifecycle from employee request submission to approval, software installation task, fulfillment, notification, dashboard reporting, and deployment evidence.

11. Conclusion and Future Enhancements

The Licensed Software Installation Request solution successfully centralizes and automates the process of requesting licensed software through the ServiceNow Service Catalog. The implementation connects structured request intake with license validation, approval, automated task creation, fulfillment, SLA monitoring, notifications, security, reporting, and deployment.

The solution reduces manual effort, improves visibility, supports controlled use of licensed software, and provides a traceable request lifecycle for employees and IT support teams.

Future Enhancements

- Add more software products and versions.
- Introduce additional approval levels for restricted or high-cost licenses.
- Expand license inventory and availability automation.
- Add more dashboard reports and analytics.
- Improve automated fulfillment and software deployment integration.
- Enhance role-based access and validation.